

Software Architecture

SE6105

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(BSc Honours in Software Engineering)

SE6105 – Course Content

- Topic 01: Basic concepts & principles about software architecture
- Topic 02: Introduction to Software Architectural pattern
- Topic 03: ADL
- Topic 04: 4+1 Architecture
- Topic 05: Practical approaches & methods for Create & Analyse software architecture
- Topic 06: Quality attributes of software architectures
- Topic 07: Examples in architectural design applications & case studies in software architecture (N tier architecture, SOA, Cloud, etc.)

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Quality attributes of software architectures

How do we evaluate whether an architecture is GOOD?

Motivation

You already know:

- How to design architectures
- How to represent them (ADL, 4+1)

But:

- How do we justify architectural decisions?
- How do we compare two architectures?

Definition

Quality Attributes are **Measurable properties** that determine how well an architecture **satisfies system requirements**

- Focus is NOT features
- Focus is system behavior under conditions

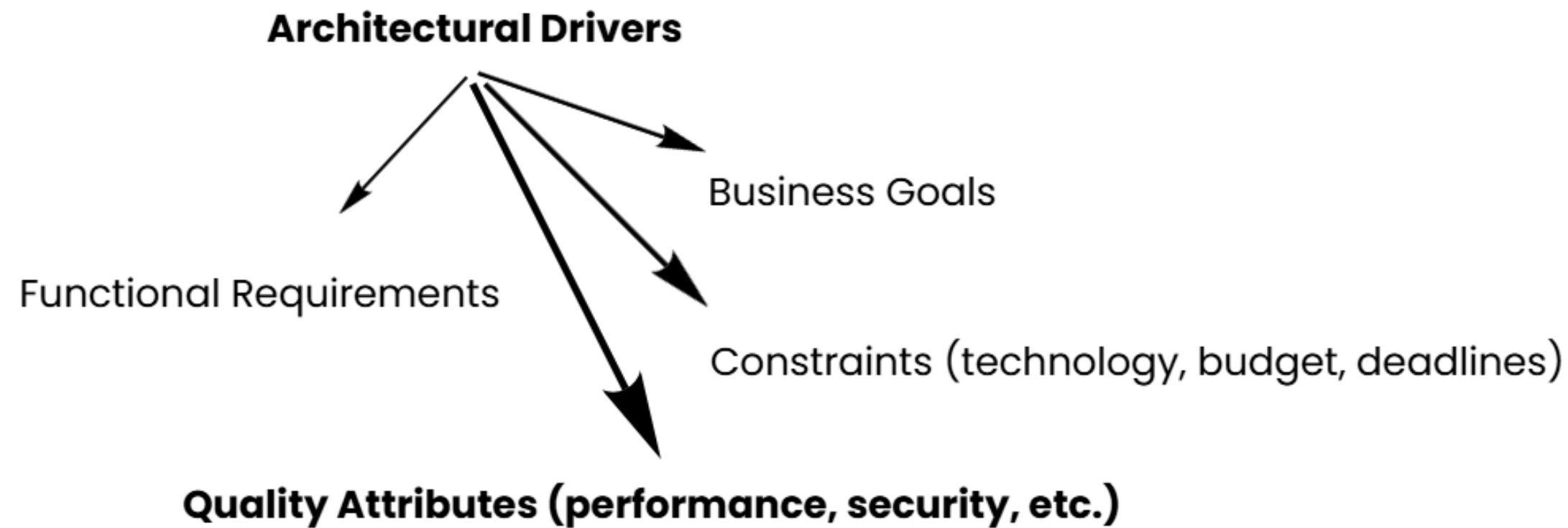
Architecture Evaluation Contexton

Quality attributes are used in:

- ATAM (Architecture Tradeoff Analysis Method)
- SAAM (Software Architecture Analysis Method)

Architecture Drivers

Architectural Drivers



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SE6103

10

Quality attributes have the strongest influence

Quality Attribute Scenarios

Structure:

- Source (who triggers)
- Stimulus (what happens)
- Environment (when)
- Artifact (affected part)
- Response (system reaction)
- Response Measure (how well)

Quality Attribute Scenarios

Structured way to define quality requirements

Component	Example
Source	User
Stimulus	1000 concurrent requests
Environment	Peak hours
Artifact	Web server
Response	Process requests
Response measure	< 2 seconds response time

Key Quality Attributes (Architecture-Oriented)

- 1. Performance**
- 2. Scalability**
- 3. Availability**
- 4. Modifiability**
- 5. Security**
- 6. Interoperability**

Trade-Off Analysis

- Performance vs Security
- Modifiability vs Performance
- Availability vs Cost

Architectural Tactics

Design decisions to achieve quality attributes

Tactic	How to achieve
Performance	caching, concurrency
Security	authentication, encryption
Availability	redundancy

Quality Attributes & Patterns

Pattern	Supports
Layered	Modifiability
Microservices	Scalability
Event-driven	Performance
Client-Server	Distribution

Mapping to 4+1 Views

View	Related Attributes
Logical	Modifiability
Process	Performance
Development	Maintainability
Physical	Availability
Scenarios	All

Real Architecture Example

Online Banking System

Attribute	Architectural Decision
Security	Multi-layer auth
Availability	Redundant servers
Performance	Caching
Scalability	Microservices

Scenario: Ride Sharing App

- What are top 3 quality attributes?
- What architecture decisions support them?

Common Mistakes (Architecture Level)

- Ignoring quality attributes early
- No measurable scenarios
- Over-designing for all attributes
- Not considering trade-offs

Key Takeaways

- Quality attributes drive architecture decisions
- Use scenarios to define them
- Use tactics & patterns to achieve them
- Always consider trade-offs

Quality Attributes

Thank You!